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Experiment no. 03

Aim: To deploy multi-service application using Docker compose

LO4: To automate provisioning, configuration, container orchestration, and cloud resource management using Ansible, Docker, Kubernetes, and Terraform.

Theory:

Docker Compose is a tool provided by Docker that is used to define and run multi-container applications using a single YAML configuration file (docker-compose.yml). Instead of creating and starting each container separately using individual docker run commands, Docker Compose allows all the services required by an application — such as a web server, a database, and supporting tools — to be defined together in one file, and started, stopped, or managed together using a single command such as docker compose up. This directly supports the aim of this experiment: deploying a multi-service application as one unit rather than as separate, manually managed containers.

In real-world applications, a single container is rarely sufficient — a typical application may require a web server, a database, a caching layer, and other supporting services running together and communicating with each other. Managing each of these manually with individual docker run commands is tedious and error-prone.

Docker Compose solves this by allowing all services, along with their images, environment variables, ports, volumes, and dependencies, to be defined declaratively in the dockercompose.yml file. A single command, docker compose up, then reads this file and creates, configures, and starts all the defined services together, connected on a common Composemanaged network so they can communicate with each other using their service names as hostnames.

Key concepts used in this experiment:

Service: A single container definition within the compose file (e.g. web, database, phpmyadmin). Each service specifies the image to use, ports to expose, environment variables, and other configuration.

Image: The base image each service is built from — nginx for the web server, mysql:8.0 for the database, and phpmyadmin/phpmyadmin for the database admin UI.

Ports mapping ("host:container"): Maps a port on the host machine to a port inside the container, e.g. "8080:80" exposes the container's port 80 on the host's port 8080.

Environment variables: Used to configure a service at startup without modifying the image itself — here they set the MySQL root password, database name, and user credentials, and tell phpMyAdmin which database host to connect to.

Volumes: Named, persistent storage (mysql-data) that keeps database data intact even if the container is stopped, removed, or recreated.

depends_on: Specifies that one service depends on another, influencing the order in which containers are started (phpmyadmin is started after database).

This experiment uses three services instead of the minimum two, to demonstrate a more realistic multi-service setup: an nginx web server, a MySQL database, and a phpMyAdmin instance for managing that database through a browser-based interface — showing how Compose handles inter-service communication (phpMyAdmin connecting to the database service purely by its service name, "database").

DOCKER COMPOSE FILE (docker-compose.yml)

The following configuration was created inside the compose-demo directory and saved as dockercompose.yml:

```
version: '3.8'
  services:    web:
image: nginx
container_name: webserver
ports:       - "8090:80"
    database:     image:
mysql:8.0     container_name:
mysqlserver     environment:
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_DATABASE: studentdb
      MYSQL_USER: student
MYSQL_PASSWORD: student123     ports:
- "3307:3306"     volumes:
- mysql-data:/var/lib/mysql
    phpmyadmin:     image:
phpmyadmin/phpmyadmin
container_name: phpmyadmin
environment:       PMA_HOST:
database
      PMA_USER: student
PMA_PASSWORD: student123
ports:       - "8081:80"
depends_on:
- database

volumes:   mysql-
data:
```

PROCEDURE

The following steps were carried out to set up and deploy the application:

#	Step / Command	Description
1	`mkdir compose-demo`	Creates a new project directory for this experiment.
2	`cd compose-demo`	Moves into the newly created directory.

3	`(create docker-compose.yml)`	The compose file is created using Notepad (saved as "All Files" so it is not saved with a .txt extension) inside the compose-demo folder.
4	`dir`	Lists the contents of the folder to confirm dockercompose.yml was saved correctly.
5	`docker compose up -d`	Reads docker-compose.yml and creates + starts all three services (webserver, mysqlserver, phpmyadmin) in detached (background) mode.

How the services connect

The web service (nginx) is accessible from the host browser at http://localhost:8090.

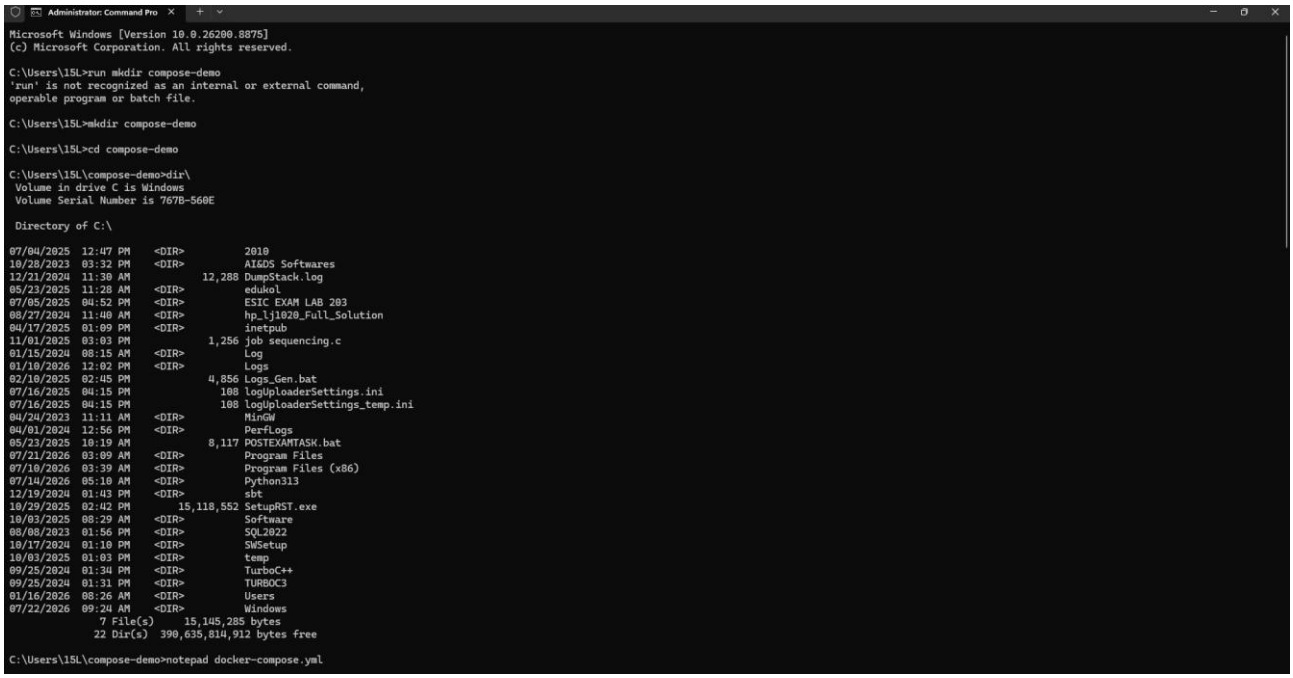
The database service (MySQL) is accessible on host port 3307, with credentials and database name set via environment variables, and its data persisted in the mysql-data volume.

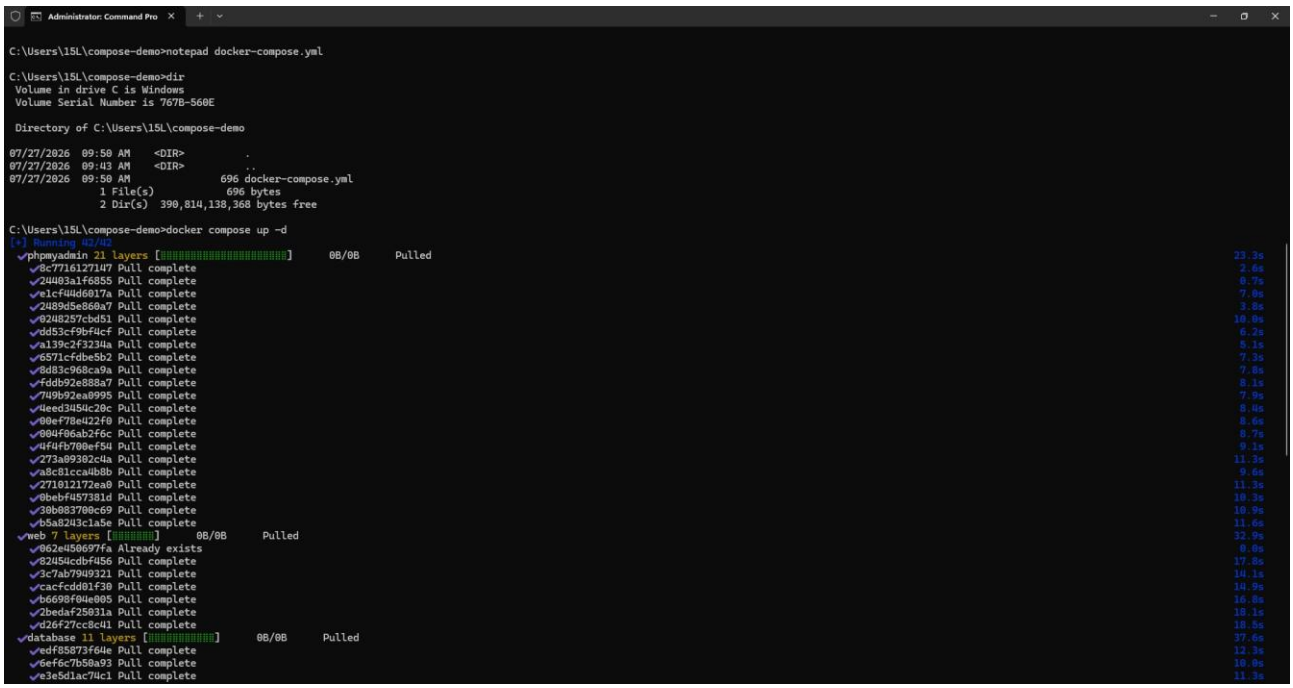
The phpmyadmin service is accessible at http://localhost:8081, and connects internally to the database service using the hostname "database" (its Compose service name) rather than an IP address — this works because Compose automatically places all services on the same internal network.

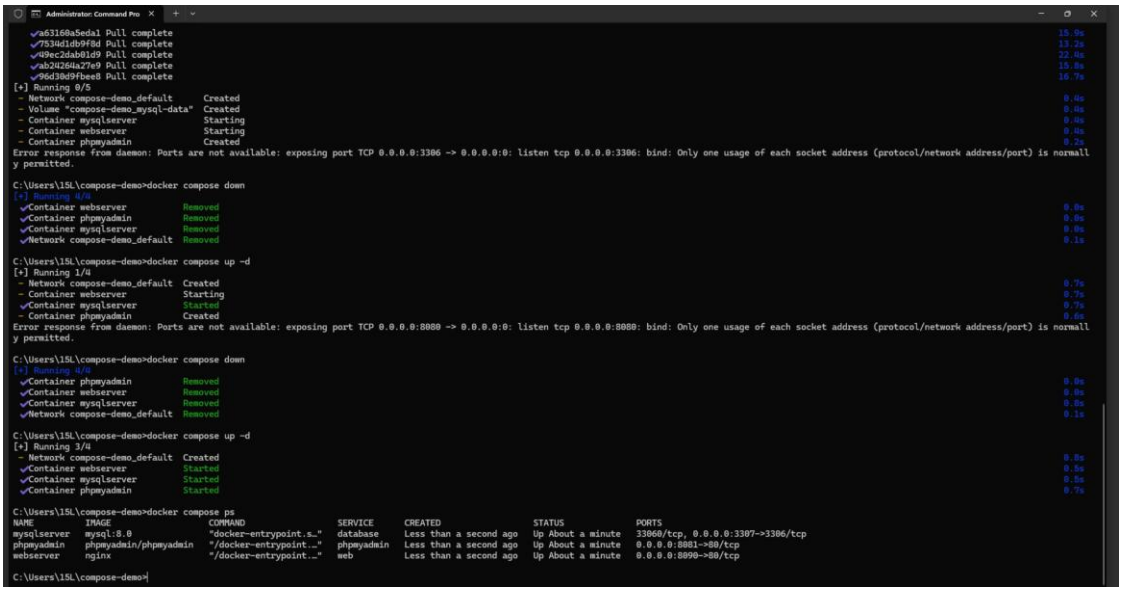
Note

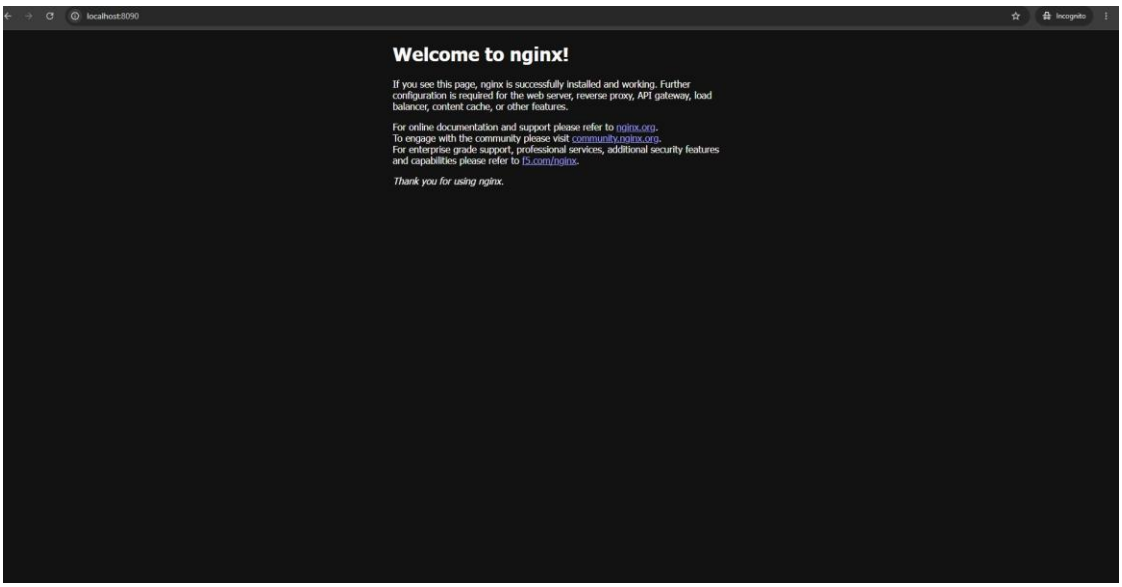
During deployment, docker compose up -d initially failed because the default host ports (3306 and 8080) specified in the file were already in use on the host machine. This was resolved by changing only the host-side port numbers to 3307 and 8090 respectively in docker-compose.yml, without altering the container-side ports, after which docker compose down followed by docker compose up -d started all three services successfully.

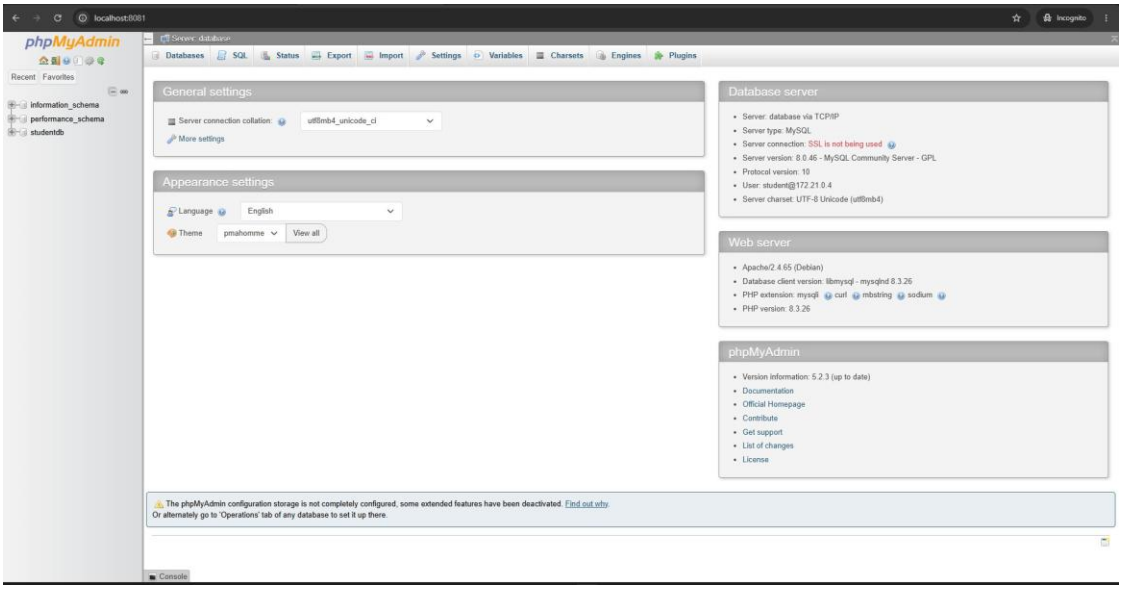
OBSERVATIONS / SCREENSHOTS




Administrator: Command Pro
C:\Users\15L\compose-demo>notepad docker-compose.yml
C:\Users\15L\compose-demo>dir
Volume in drive C is Windows
Volume Serial Number is 767B-560E
Directory of C:\Users\15L\compose-demo
07/27/2026 09:50 AM <DIR> .
07/27/2026 09:43 AM <DIR> ..
07/27/2026 09:50 AM 696 docker-compose.yml
1 File(s) 696 bytes
2 Dir(s) 390,814,138,368 bytes free
C:\Users\15L\compose-demo>docker compose up -d
[+] Running 42/42
phpmyadmin 21 layers [#####################] 0B/0B Pulled 23.3s
8c7716127147 Pull complete 2.6s
24403a1f6855 Pull complete 0.7s
e1cf44d6017a Pull complete 7.0s
2489d5e860a7 Pull complete 3.8s
0248257cbd51 Pull complete 10.0s
dd53cf9bf4cf Pull complete 6.2s
a139c2f3234a Pull complete 5.1s
6571cfdbe5b2 Pull complete 7.3s
8d83c968ca9a Pull complete 7.8s
fddb92e888a7 Pull complete 8.1s
749b92ea0995 Pull complete 7.9s
4eed3454c20c Pull complete 8.4s
00ef78e422f0 Pull complete 8.6s
004f06ab2f6c Pull complete 8.7s
4f4fb700ef54 Pull complete 9.1s
273a09302c4a Pull complete 11.3s
a8c81cca0b8b Pull complete 9.6s
271012172ea0 Pull complete 11.3s
0bebf457381d Pull complete 10.3s
30b083700c69 Pull complete 10.9s
b5a8243c1a5e Pull complete 11.6s
web 7 layers [########] 0B/0B Pulled 32.9s
062e450697fa Already exists 0.0s
82454cdbf456 Pull complete 17.8s
3c7ab7949321 Pull complete 14.1s
cacfcdd01f30 Pull complete 14.9s
b6698f04e005 Pull complete 16.8s
2bedaf25031a Pull complete 18.1s
d26f27cc8c41 Pull complete 18.5s
database 11 layers [###########] 0B/0B Pulled 37.6s
edf85873f64e Pull complete 12.3s
6ef6c7b50a93 Pull complete 10.0s
e3e5d1ac74c1 Pull complete 11.3s


Administrator: Command Pro
a63160a5eda1 Pull complete 15.9s
7534d1db9f8d Pull complete 13.2s
49ec2dab01d9 Pull complete 22.4s
ab242642a27e9 Pull complete 15.8s
96d30d9f1ee8 Pull complete 16.7s
[+] Running 0/5
Network compose-demo_default Created 0.4s
Volume "compose-demo_mysql-data" Created 0.4s
Container mysqlserver Starting 0.4s
Container webserver Starting 0.4s
Container phpmyadmin Created 0.2s
Error response from daemon: Ports are not available: exposing port TCP 0.0.0.0:3306 -> 0.0.0.0:0: listen tcp 0.0.0.0:3306: bind: Only one usage of each socket address (protocol/network address/port) is normally permitted.
C:\Users\15L\compose-demo>docker compose down
[+] Running 4/4
Container webserver Removed 0.0s
Container phpmyadmin Removed 0.0s
Container mysqlserver Removed 0.0s
Network compose-demo_default Removed 0.1s
C:\Users\15L\compose-demo>docker compose up -d
[+] Running 1/4
Network compose-demo_default Created 0.7s
Container webserver Starting 0.7s
Container mysqlserver Started 0.7s
Container phpmyadmin Created 0.6s
Error response from daemon: Ports are not available: exposing port TCP 0.0.0.0:8080 -> 0.0.0.0:0: listen tcp 0.0.0.0:8080: bind: Only one usage of each socket address (protocol/network address/port) is normally permitted.
C:\Users\15L\compose-demo>docker compose down
[+] Running 4/4
Container phpmyadmin Removed 0.0s
Container webserver Removed 0.0s
Container mysqlserver Removed 0.0s
Network compose-demo_default Removed 0.1s
C:\Users\15L\compose-demo>docker compose up -d
[+] Running 3/4
Network compose-demo_default Created 0.0s
Container webserver Started 0.5s
Container mysqlserver Started 0.5s
Container phpmyadmin Started 0.7s
C:\Users\15L\compose-demo>docker compose ps
NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
mysqlserver mysql:8.0 "docker-entrypoint.s…" database Less than a second ago Up About a minute 33060/tcp, 0.0.0.0:3307->3306/tcp
phpmyadmin phpmyadmin/phpmyadmin "/docker-entrypoint.…" phpmyadmin Less than a second ago Up About a minute 0.0.0.0:8081->80/tcp
webserver nginx "/docker-entrypoint.…" web Less than a second ago Up About a minute 0.0.0.0:8090->80/tcp
C:\Users\15L\compose-demo>


localhost:8090
Incognito
Welcome to nginx!
If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.
For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.
Thank you for using nginx.



DOCKER COMPOSE COMMANDS USED

The following Docker Compose commands were used to manage the lifecycle of the multiservice application:

#	Command	Description
1	`docker compose up -d`	Starts all services defined in the compose file, in detached mode.
2	`docker compose ps`	Shows the status of all running services defined in the compose file.
3	`docker compose logs`	Displays combined logs from all services, useful for debugging.
4	`docker compose images`	Lists the images used by each service in the compose file.
5	`docker compose config`	Validates the syntax of the compose file and displays the fully resolved configuration.
6	`docker compose stop`	Stops all running services without removing the containers.
7	`docker compose start`	Starts services that were previously stopped.
8	`docker compose restart`	Restarts all containers defined in the compose file.
9	`docker compose down`	Stops and removes all containers and the default network, but keeps named volumes.
10	`docker compose down -v`	Stops and removes all containers, networks, AND named volumes (deletes database data) — run this last.

CONCLUSION

Through this experiment, a multi-service application consisting of a web server, a database, and a database administration tool was successfully deployed using Docker Compose. Instead of creating and managing each container individually, a single docker-compose.yml file was used to declaratively define all three services, their images, ports, environment variables, and persistent storage. The docker compose up -d command started all services together with a single instruction, and docker compose ps confirmed that all three containers were running and correctly networked. This experiment demonstrates the core advantage of Docker Compose over standalone Docker commands: it simplifies the deployment and orchestration of applications composed of multiple interdependent services, allows services to communicate using simple service names instead of IP addresses, and makes the entire application's configuration reproducible and versioncontrollable through a single file.